AudioLLM-D: Fine-tuning Audio LLM for Dementia speech

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1 Literature Review

Speech offers a compelling non-invasive biomarker: as cognitive decline progresses, patients exhibit characteristic disfluencies including filled pauses, word-finding hesitations, reduced lexical richness, and fragmented syntax that worsen measurably and span acoustic, phonological, and discourse levels [17]. The landscape of AI research for dementia can be broadly organized into three evolving categories: acoustic signal processing, which uses audio features as primary biomarkers of cognitive decline; specialized speech-to-text (ASR), aiming to preserve the linguistic disfluencies that standard "fluency-biased" models suppress; and text-based reasoning, leveraging Large Language Models (LLMs) to analyze conversation transcripts for signs of cognitive decline. While each field has made significant progress, they remain largely isolated.

1.1 Acoustic Signal Processing

Early approaches relied on hand-crafted acoustic features (fundamental frequency, jitter, pause duration, etc.) and linguistic measures (type-token ratio, mean length of utterance) [7] [6]. The ADReSS Challenge [9] established the first age- and gender-balanced benchmark, showing that linguistic features from manual transcripts outperform acoustic features alone. The follow-up ADReSSo Challenge [10] removed manual transcripts, and Qiao et al. [11] showed that combining ASR-derived disfluency features with fine-tuned BERT [5]/ERNIE [14] ensembles achieves strong classification accuracy.

1.2 Specialized Clinical Speech Recognition

The primary bottleneck in automated dementia screening is the "fluency bias" of standard Automatic Speech Recognition (ASR): Models optimized to minimize Word Error Rate (WER) suppress non-lexical disfluencies, filtering out the most informative clinical markers. Early specialized systems addressed this through targeted architectural design - [18] developed a Conformer-based ASR system on DementiaBank [2], with elderly speaker adaptation, substantially reducing WER and improving downstream Alzheimer's Disease detection. The shift toward foundation speech models, particularly Whisper [12], subsequently raised the performance ceiling by preserving certain paralinguistic cues without requiring manual feature engineering [8]. WhisperD [1] fine-tuned Whisper on dementia corpora, improving the Filler Inclusion Rate (FIR) for lexical fillers such as *um*, *uh* etc. However, a critical gap remains: current systems lack a mechanism to discretely tokenize temporal silence. The duration of silent pauses, a hallmark of anomia, is rarely treated as a primary data token, limiting the model's ability to provide a complete temporal profile of the patient's cognitive struggle.

1.3 LLM reasoning based dementia diagnosis

Recent work has shifted from classification toward interactive LLM-based screening agents. A review by [16] highlights the clinical potential of LLMs for symptom tracking while cautioning against hallucination risks. The DemMA framework [13] uses a multi-turn agent with an "Observe-Reason-Action" cycle to guide follow-up questioning. [15] and [3] further identify reasoning-augmented prompting and token-level fine-tuning as effective adaptation strategies. However, these dementia-focused frameworks remain text-centric, operating on transcripts alone. While Audio-Reasoner [20] demonstrates that audio-specific Chain-of-Thought [19] training enables reasoning over acoustic signals, this approach has not been applied to dementia screening. Our work bridges this gap by adapting Qwen2-Audio [4] to dementia speech, enabling transcription that preserves both lexical disfluencies and temporal silence as discrete tokens, with downstream reasoning evaluated via QA pairs.

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